

CAPSTONE PROJECT

USECASE 2 – Exploratory Analysis & Visualization of Crime

Name : Mamidoju.Vishal

Email : vishalmamidoju5@gmail.com

Objective: Analysis trends and patterns using Pandas,Numpy, Matplotlib and Seaborn.

Tasks:

1.Crime Trend Over Years:

- Plot total number of crimes per year.
- Visual interpretation: Is the crime rate rising or decreasing?

Code:

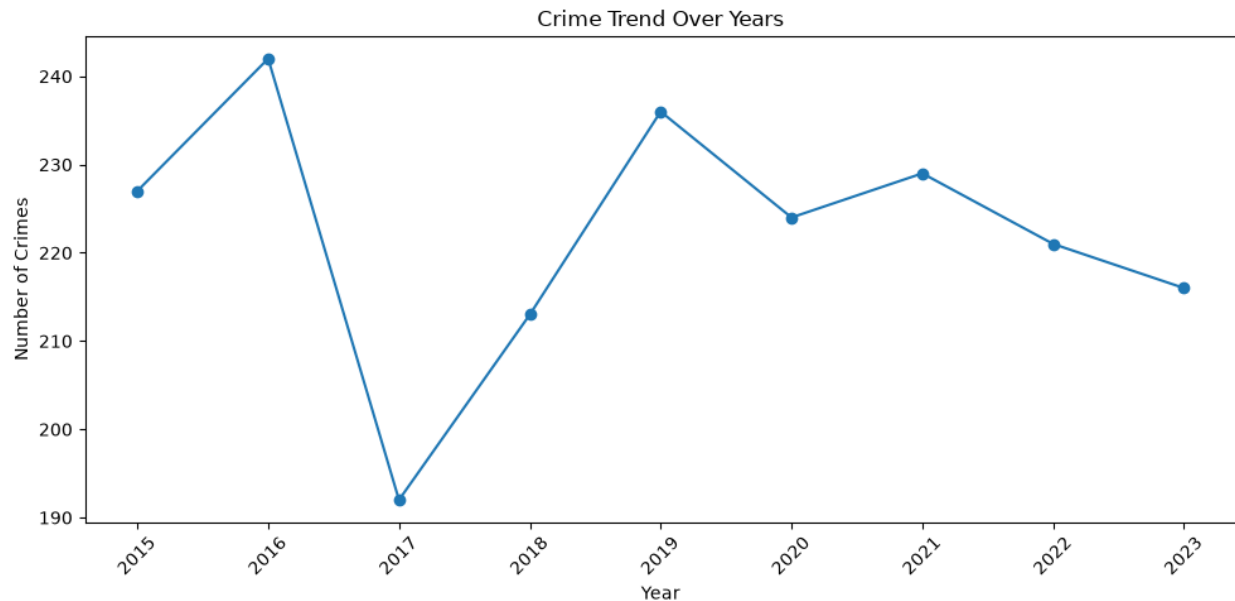
```
# ===== 1. CRIME TREND =====

yearly_crimes = df.groupby("crime_year").size()

print("\n===== CRIME TREND BY YEAR =====")
print(yearly_crimes)

plt.figure(figsize=(10, 5))
plt.plot(yearly_crimes.index, yearly_crimes.values, marker="o")
plt.title("Crime Trend Over Years")
plt.xlabel("Year")
plt.ylabel("Number of Crimes")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "crime_trend.png"), bbox_inches="tight")
plt.close()
```

Output:



2. Crime Distribution by Category:

- Bar chart of top 10 crime categories (Primary Type).
- Calculate counts and percentage for each.

Code:

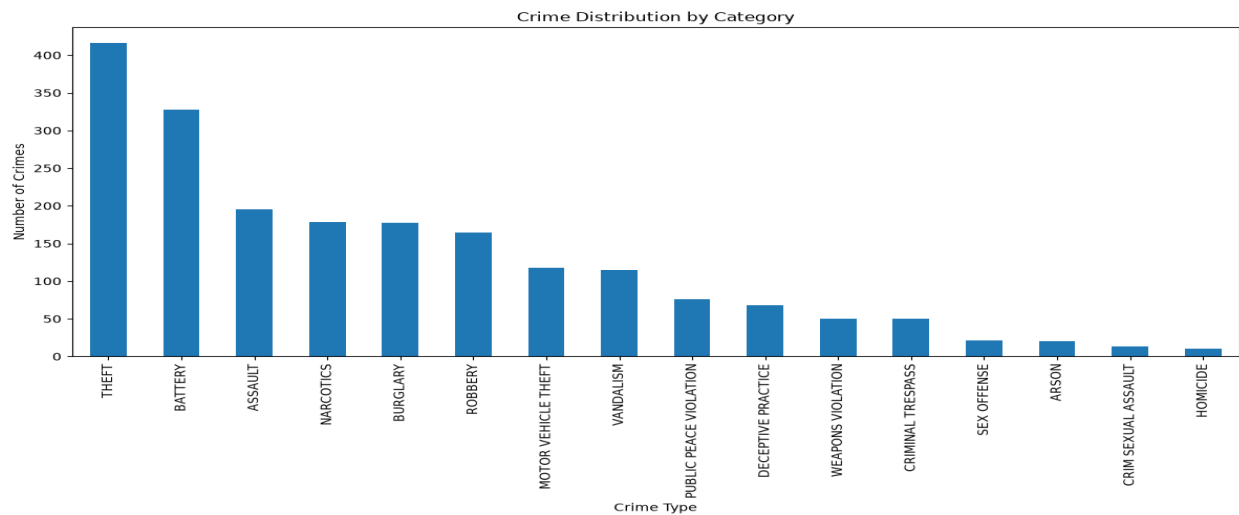
```
# ===== 2. CRIME DISTRIBUTION =====

crime_category = df["primary_type"].value_counts()

print("\n===== CRIME DISTRIBUTION =====")
print(crime_category)

plt.figure(figsize=(12, 7))
crime_category.plot(kind="bar")
plt.title("Crime Distribution by Category")
plt.xlabel("Crime Type")
plt.ylabel("Number of Crimes")
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "crime_distribution.png"), bbox_inches="tight")
plt.close()
```

Output:



3.Arrests and Crime Outcomes:

- What percentage of crimes result in arrest?
- `arrest_rate = df['Arrest'].mean() * 100`

Code:

```
# ===== 4. ARREST PERCENTAGE =====

top_10_names = top_10_crimes.index
top_10_data = df[df["primary_type"].isin(top_10_names)]

arrest_percentage = (
    top_10_data.groupby("primary_type")["arrest"]
    .mean()
    .mul(100)
    .sort_values(ascending=False)
)

print("\n===== ARREST PERCENTAGE =====")
print(arrest_percentage)

plt.figure(figsize=(10, 6))
arrest_percentage.plot(kind="bar")
plt.title("Arrest Percentage for Top 10 Crime Types")
plt.xlabel("Crime Type")
plt.ylabel("Arrest Percentage (%)")
plt.xticks(rotation=60)
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "arrest_percentage.png"), bbox_inches="tight")
plt.close()
```

Output:

```
===== ARREST PERCENTAGE =====
primary_type
NARCOTICS                56.741573
BATTERY                  41.768293
ROBBERY                  38.787879
VANDALISM                33.043478
ASSAULT                  29.743590
DECEPTIVE PRACTICE     27.941176
PUBLIC PEACE VIOLATION   26.315789
MOTOR VEHICLE THEFT      16.101695
THEFT                    14.903846
BURGLARY                 12.994350
Name: arrest, dtype: float64
```

4.Heatmap of Crime by Month and Day of Week

- Use Seaborn heatmap of crime frequency pivoted by Month vs DayOfWeek.

Code:

```
# ===== 5. CRIME HEATMAP =====

day_order = [
    "Monday", "Tuesday", "Wednesday", "Thursday",
    "Friday", "Saturday", "Sunday"
]

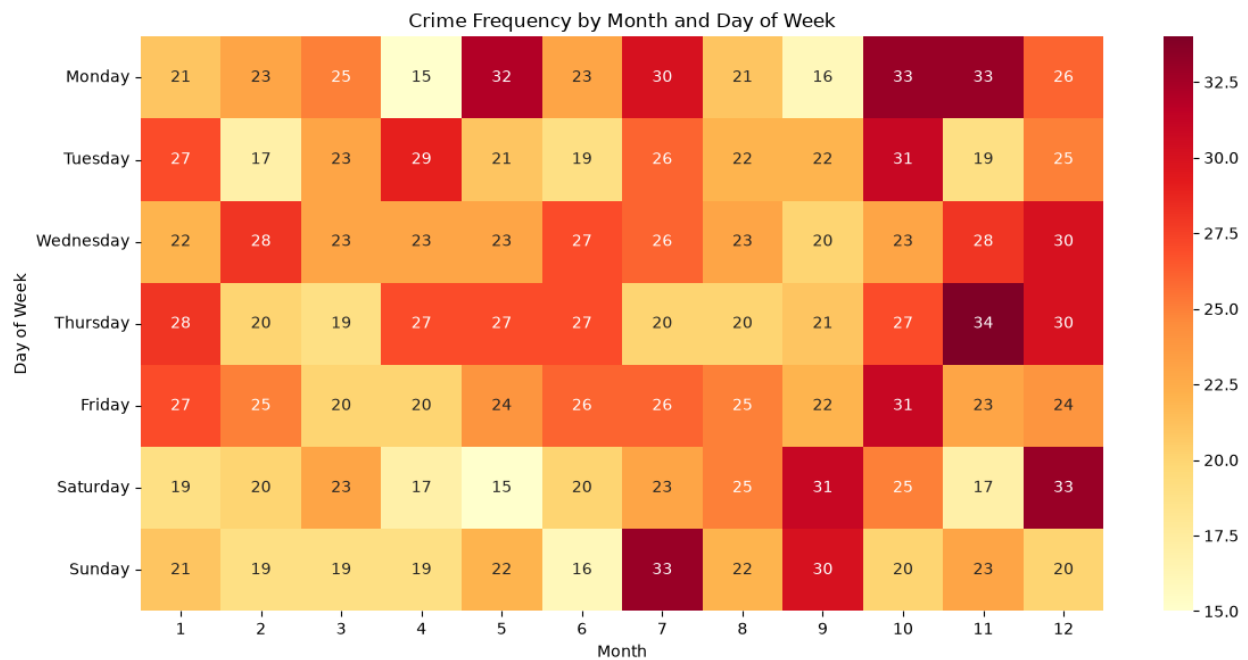
df["day_of_week"] = pd.Categorical(
    df["day_of_week"],
    categories=day_order,
    ordered=True
)

heatmap_data = pd.crosstab(df["day_of_week"], df["crime_month"])

print("\n===== CRIME HEATMAP DATA =====")
print(heatmap_data)

plt.figure(figsize=(12, 6))
sns.heatmap(heatmap_data, annot=True, fmt="d", cmap="YlOrRd")
plt.title("Crime Frequency by Month and Day of Week")
plt.xlabel("Month")
plt.ylabel("Day of Week")
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "crime_heatmap.png"), bbox_inches="tight")
plt.close()
```

Output:



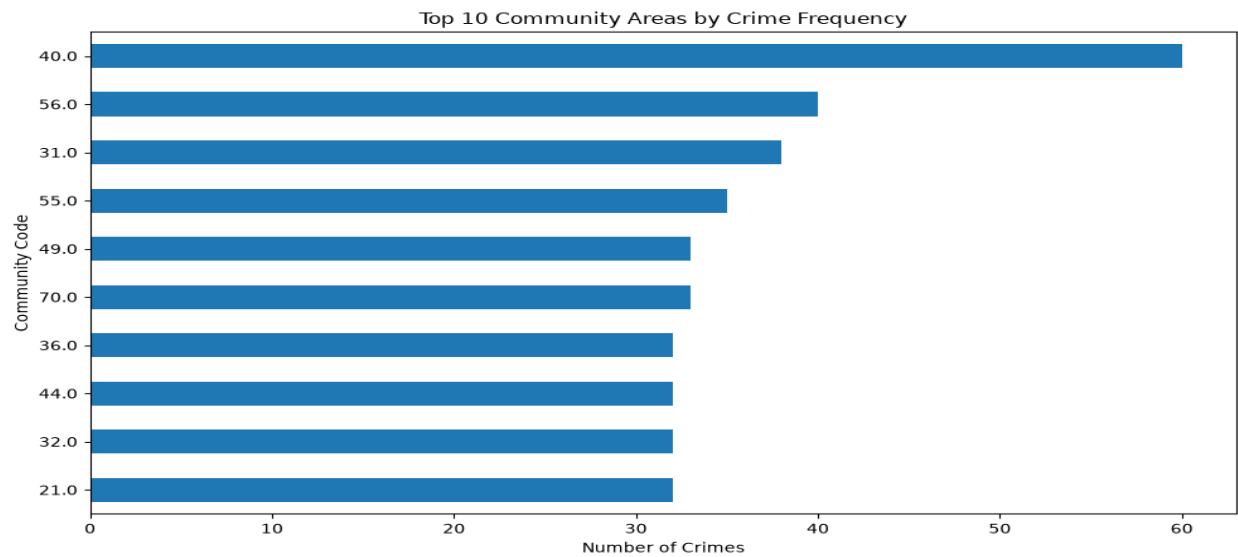
5.Top Community Areas:

- List top 10 community areas with highest crime counts.
- Plot on a bar chart.

Code:

```
# ===== 6. TOP COMMUNITY AREAS =====  
  
community_crimes = df["community_code"].value_counts().head(10)  
  
print("\n===== TOP 10 COMMUNITY AREAS =====")  
print(community_crimes)  
  
plt.figure(figsize=(10, 6))  
community_crimes.sort_values().plot(kind="barh")  
plt.title("Top 10 Community Areas by Crime Frequency")  
plt.xlabel("Number of Crimes")  
plt.ylabel("Community Code")  
plt.tight_layout()  
plt.savefig(os.path.join(CHART_DIR, "top_community_areas.png"), bbox_inches="tight")  
plt.close()
```

Output:



Questions

- Which crime category is most frequent?

Code:

```
# ===== 3. TOP 10 CRIMES =====

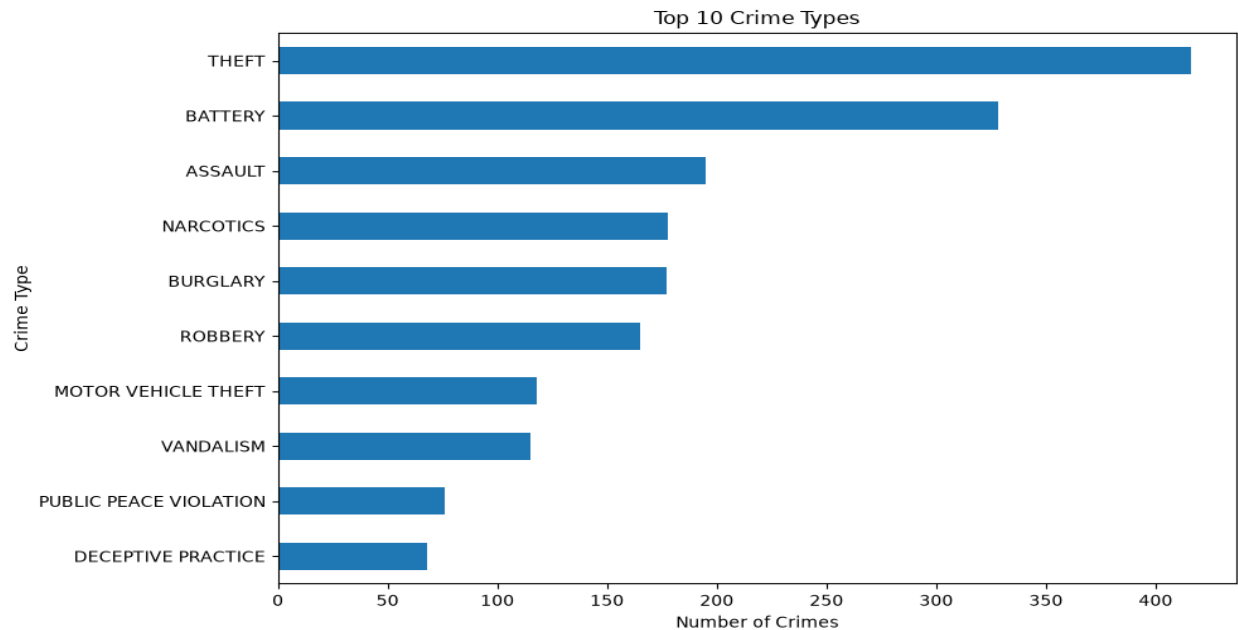
top_10_crimes = df["primary_type"].value_counts().head(10)

print("\n===== TOP 10 CRIME TYPES =====")
print(top_10_crimes)

plt.figure(figsize=(10, 6))
top_10_crimes.sort_values().plot(kind="barh")
plt.title("Top 10 Crime Types")
plt.xlabel("Number of Crimes")
plt.ylabel("Crime Type")
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "top_10_crimes.png"), bbox_inches="tight")
plt.close()

# ===== 4. ARREST PERCENTAGE =====
```

Output:



- Is the arrest rate consistent across different years?

Code:

```
# ===== 4. ARREST PERCENTAGE =====

top_10_names = top_10_crimes.index
top_10_data = df[df["primary_type"].isin(top_10_names)]

arrest_percentage = (
    top_10_data.groupby("primary_type")["arrest"]
    .mean()
    .mul(100)
    .sort_values(ascending=False)
)

print("\n===== ARREST PERCENTAGE =====")
print(arrest_percentage)

plt.figure(figsize=(10, 6))
arrest_percentage.plot(kind="bar")
plt.title("Arrest Percentage for Top 10 Crime Types")
plt.xlabel("Crime Type")
plt.ylabel("Arrest Percentage (%)")
plt.xticks(rotation=60)
plt.tight_layout()
plt.savefig(os.path.join(CHART_DIR, "arrest_percentage.png"), bbox_inches="tight")
plt.close()
```

Output:

```
===== ARREST PERCENTAGE =====
primary_type
NARCOTICS          56.741573
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MOTOR VEHICLE THEFT 16.101695
THEFT              14.903846
BURGLARY           12.994350
Name: arrest, dtype: float64
```

- Which month has the highest crime frequency?

Code:

```
# ===== 7. MONTHLY CRIME ANALYSIS =====

monthly_crimes = df["crime_month"].value_counts().sort_index()
highest_crime_month = monthly_crimes.idxmax()
highest_crime_count = monthly_crimes.max()

print("\n===== MONTHLY CRIME FREQUENCY =====")
print(monthly_crimes)

print("\n=====")
print(f"MONTH WITH HIGHEST CRIME FREQUENCY: {highest_crime_month}")
print(f"NUMBER OF CRIMES: {highest_crime_count}")
print("=====")
```


Output:

```
===== MONTHLY CRIME FREQUENCY =====  
crime_month  
1      165  
2      152  
3      152  
4      150  
5      164  
6      158  
7      184  
8      158  
9      162  
10     190  
11     177  
12     188  
Name: count, dtype: int64  
  
=====  
MONTH WITH HIGHEST CRIME FREQUENCY: 10  
NUMBER OF CRIMES: 190  
=====
```

Insights

- The crime data was analyzed to identify trends across different years, crime categories, months, days, and community areas.
- Crime distribution and the top 10 crime types were visualized using bar charts.
- Arrest percentages were calculated for the top 10 most common crime types.
- A heatmap was used to analyze the relationship between crime frequency, months, and days of the week.
- The top 10 community areas with the highest crime frequency were identified.
- Monthly crime analysis was performed to determine the month with the highest number of recorded crimes.